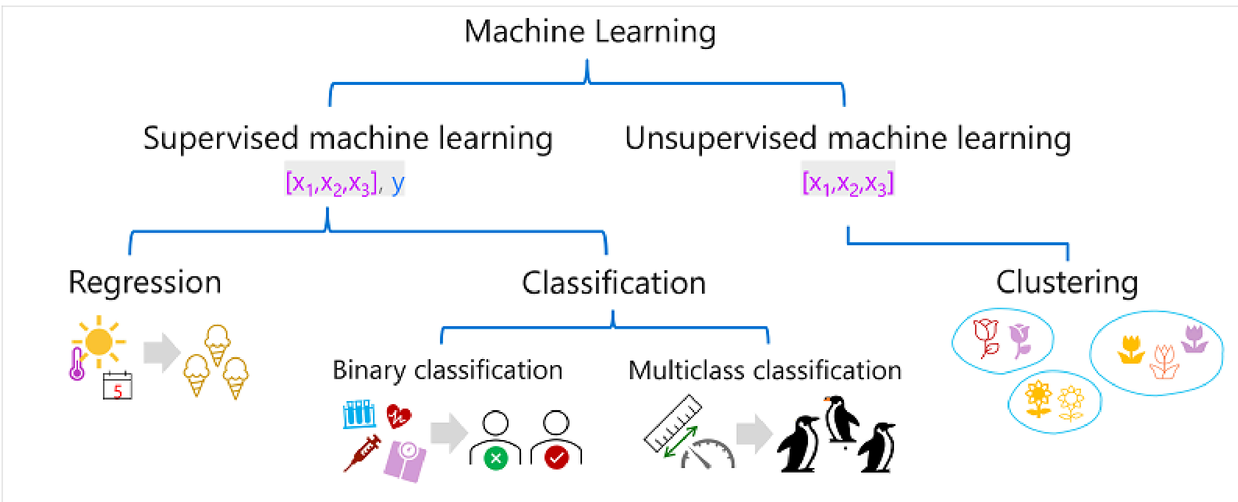
Types of machine learning

There are multiple types of machine learning, and you must apply the appropriate type depending on what you're trying to predict. A breakdown of common types of machine learning is shown in the following diagram.



Supervised machine learning

Supervised machine learning is a general term for machine learning algorithms in which the training data includes both feature values and known label values. Supervised machine learning is used to train models by determining a relationship between the features and labels in past observations, so that unknown labels can be predicted for features in future cases.

Regression

Regression is a form of supervised machine learning in which the label predicted by the model is a numeric value. For example:

- The number of ice creams sold on a given day, based on the temperature, rainfall, and windspeed.
- The selling price of a property based on its size in square feet, the number of bedrooms it contains, and socio-economic metrics for its location.
- The fuel efficiency (in miles-per-gallon) of a car based on its engine size, weight, width, height, and length.

Classification

Classification is a form of supervised machine learning in which the label represents a categorization, or class. There are two common classification scenarios.

Binary classification

In binary classification, the label determines whether the observed item is (or isn't) an instance of a specific class. Or put another way, binary classification models predict one of two mutually exclusive outcomes. For example:

- Whether a patient is at risk for diabetes based on clinical metrics like weight, age, blood glucose level, and so on.
- Whether a bank customer will default on a loan based on income, credit history, age, and other factors.
- Whether a mailing list customer will respond positively to a marketing offer based on demographic attributes and past purchases.

In all of these examples, the model predicts a binary true/false or positive/negative prediction for a single possible class.

Multiclass classification

Multiclass classification extends binary classification to predict a label that represents one of multiple possible classes. For example,

- **The species of a penguin (Adelie, Gentoo, or Chinstrap) based on its physical measurements.**
- **The genre of a movie (comedy, horror, romance, adventure, or science fiction) based on its cast, director, and budget.**

In most scenarios that involve a known set of multiple classes, multiclass classification is used to predict mutually exclusive labels. For example, a penguin can't be both a Gentoo and an Adelie. However, there are also some algorithms that you can use to train multilabel classification models, in which there may be more than one valid label for a single observation. For example, a movie could potentially be categorized as both science fiction and comedy.